From scatter to distribution function. The scatter plot in the companion figure displays individual pseudo-observation pairs (u_i, v_i)—a bivariate *density* view of the data. Point clouds like this are natural for revealing clustering, outliers, and the general shape of dependence, but they correspond to the joint *density* $h(u, v)$, not to the joint *distribution function* $H(u, v) = \Pr(U \le u,\ V \le v)$. Sklar's theorem, however, operates at the level of *distribution functions*: it decomposes the joint CDF, not the joint density. The contour lines overlaid on the scatter bridge this gap. Each contour traces out a level set of the copula CDF, $\{(u, v) : C(u, v) = c\}$, converting the point-cloud picture into an equivalent distributional picture. In essence, the scatter shows *where mass is concentrated* (a density perspective), while the contours show *how much mass is accumulated* below any point in the unit square (a CDF perspective). Understanding both views simultaneously is key to interpreting what Sklar's theorem unlocks: a clean separation of marginal behavior from dependence structure, stated entirely in the language of distribution functions.[1]

Sklar's theorem and the copula factorization. Any bivariate distribution $H(x, y)$ with continuous marginals F_X and F_Y can be uniquely decomposed as

$$H(x, y) \;=\; C\big(F_X(x),\, F_Y(y)\big), \tag{M.4}$$

where $C\colon [0, 1]^2 \to [0, 1]$ is the *copula*—a bivariate CDF with Uniform(0, 1) margins that encodes the full dependence structure between X and Y independently of the marginal scales. The accompanying figure makes this factorization visually explicit. The CDF strips on the bottom and left carry raw scores into the unit square ($u = F_X^{\mathrm{ref}}(x)$, $v = F_Y^{\mathrm{ref}}(y)$), while the density/proportion strips show how the subgroup's marginals depart from the population baseline. What remains in the interior—the bivariate pattern of the scatter—is governed entirely by the copula C.[2]

Interpreting the CDF level curves. Overlaid on the scatter are level curves of the copula CDF, $\{(u, v) : C(u, v) = c\}$ for $c = 0.1, 0.2, \dots, 0.9$. Each curve bounds a region of the unit square such that $C(u, v) = c$ means that, under the fitted dependence model, a fraction c of the population has *both* a prior pseudo-observation $\le u$ and a current pseudo-observation $\le v$. Equivalently, $C(u, v) = \Pr(U \le u,\ V \le v)$. The spacing and curvature of the contours reflect the strength and shape of dependence: under independence ($C = \Pi$, the product copula) these curves would be rectangular hyperbolas; with positive dependence they "bow" toward the main diagonal, concentrating probability mass along it.[3]

Empirical versus parametric copula. Two sets of contours are displayed: *dashed grey* lines show the empirical Bernstein copula estimated non-parametrically from the linked data, and *solid colored* lines show the best-fitting parametric copula (a t-copula with df = 25.2). The close agreement between the two confirms that the parametric family captures the essential dependence features of the data, including symmetric tail dependence ($\lambda = 0.227$) and overall rank correlation ($\tau = 0.698$, $\rho_s = 0.890$). Where the two sets of contours diverge—typically in the tails—the empirical contours reveal local structure that the parametric model smooths over.[4]

From copula to growth percentiles. Given the copula C, the conditional distribution of current achievement given prior achievement follows immediately:

$$F_{V|U}(v \mid u) \;=\; \frac{\partial\, C(u, v)}{\partial\, u}, \tag{M.5}$$

which yields a student-level growth percentile for each observed pair (u_i, v_i). The CDF level curves in the figure are thus directly related to growth percentiles: fixing u and reading off the v-values at each contour level traces out the conditional quantile function. Only *stayers*—students measured at both occasions—contribute paired observations to the empirical copula; leavers and entrants appear solely in the marginal strips.[5]

Sensitivity to copula choice. The primary visual comparison between the empirical and parametric contours highlights the robustness (or fragility) of model-based growth percentiles. If the contour sets are nearly coincident, as in the present case (ΔAIC = 210.7 vs. the next-best family), the parametric copula is an adequate summary and the resulting growth percentiles are stable. Visible divergence would signal that percentile estimates near the affected quantile regions are sensitive to the copula specification—a key diagnostic for growth-model calibration.